

# **VESICULOBULLOUS AND ULCERATIVE LESIONS OF ORAL CAVITY**

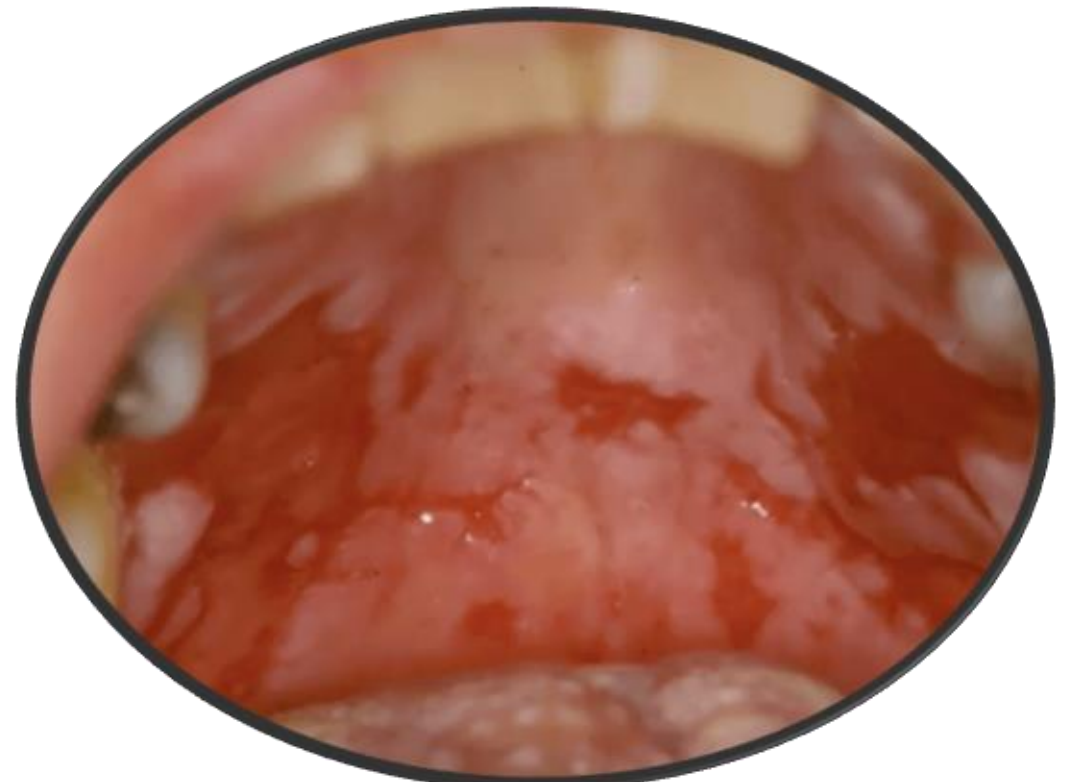
## **LEARNING OUTCOMES**

- **By the end of this chapter students will be able to:**
  1. Describe the etiopathogenesis, complications and management of common vesiculobullous and ulcerative diseases affecting mouth.
  2. Explain the clinicopathologic features of Pemphigus, Pemphigoid and Aphthous Stomatitis.
  3. Correlate the clinical and microscopic features of common vesiculobullous and ulcerative diseases to arrive at diagnosis.

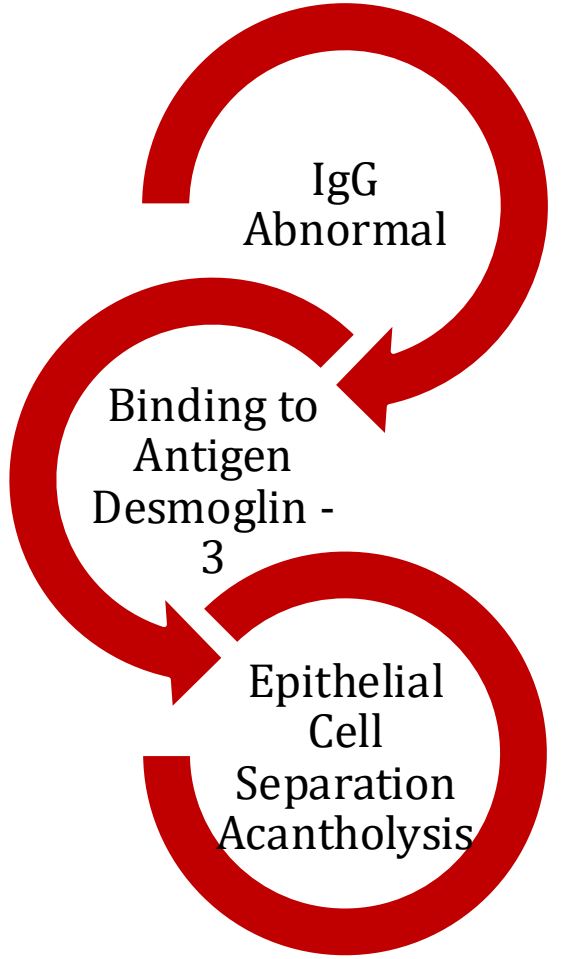
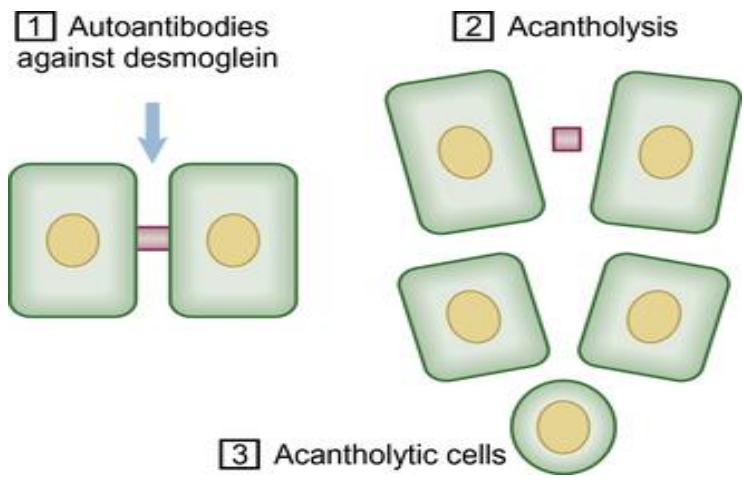
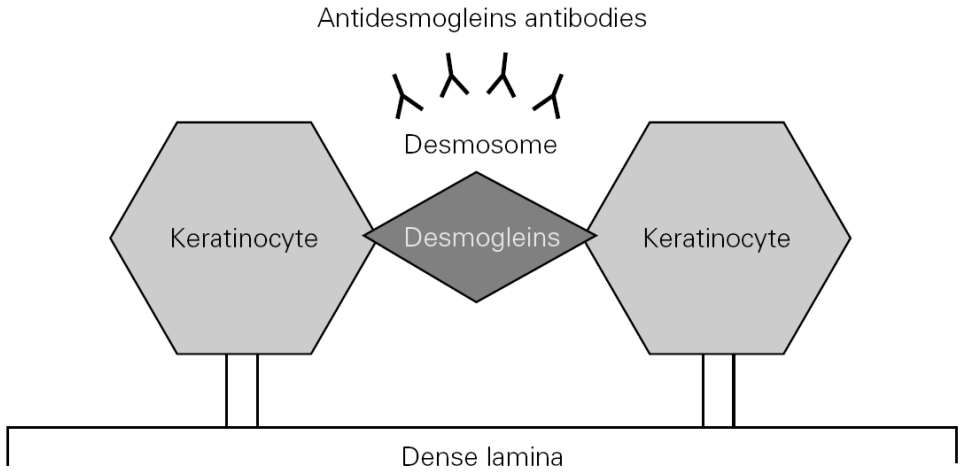
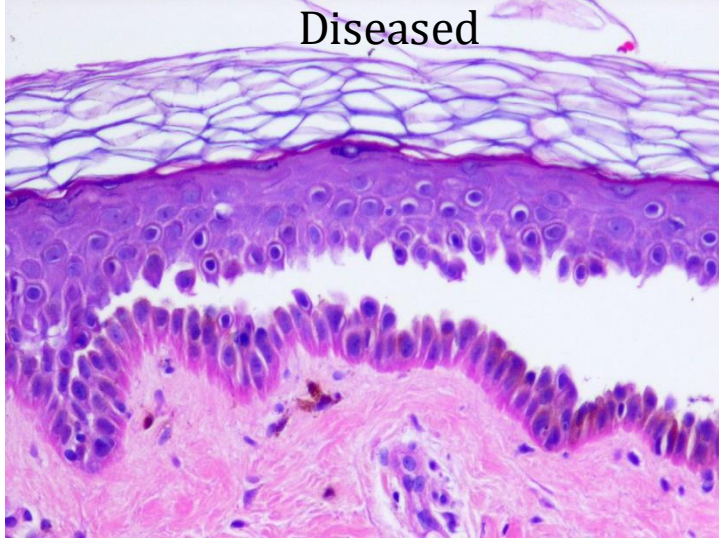
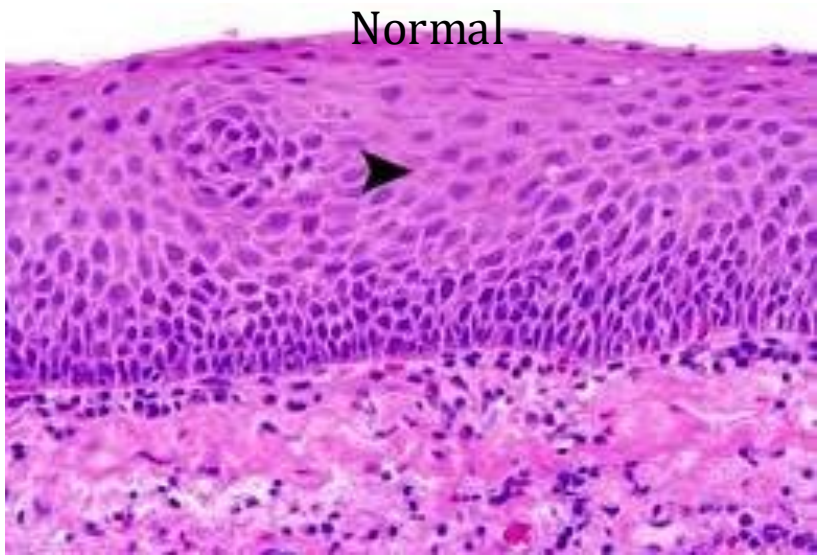
# PEMPHIGUS

## **Pemphix: Blister**

Autoimmune Muco-cutaneous disease characterized by Intraepithelial blister formation



# ETIOPATHOGENESIS



## CLINICAL FEATURES

- Wide spread painful Ulcers developed from Vesicles/Bullae – Skin and/or Mucosa
- Fluid loss and electrolyte imbalance
- Secondary infections
- Oral cavity is common location for first appearance; “first to show last to go”



**Nikolsky Sign** : Dislodging of epidermis by lateral finger pressure in the vicinity of lesions, which leads to an erosion.

Shearing stresses on normal skin can cause new erosions to form

## CLINICAL TYPES



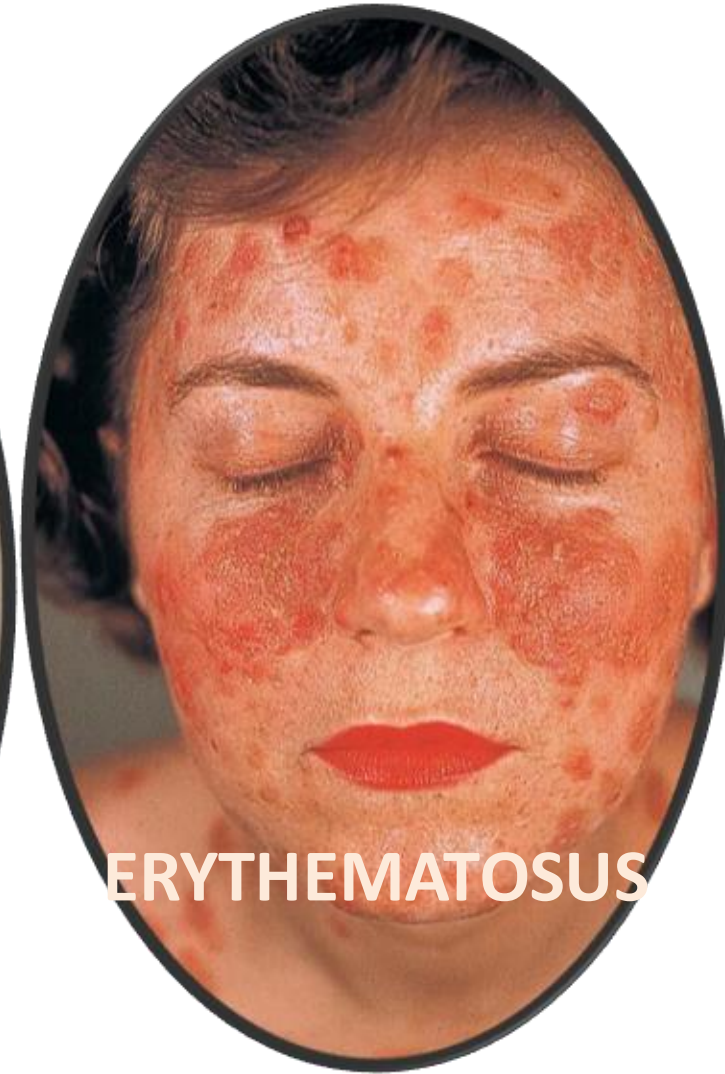
VULGARIS



FOLIACEOUS



VEGETANS



ERYTHEMATOSUS

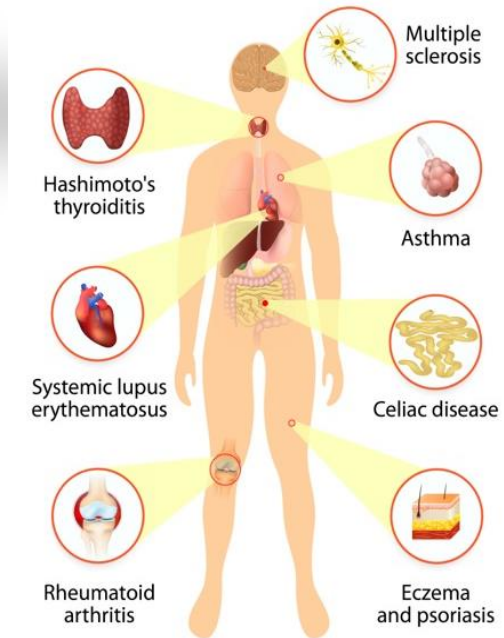
## Drug Induced



## Para Neoplastic



**Associated with Other  
Auto Immune Diseases**

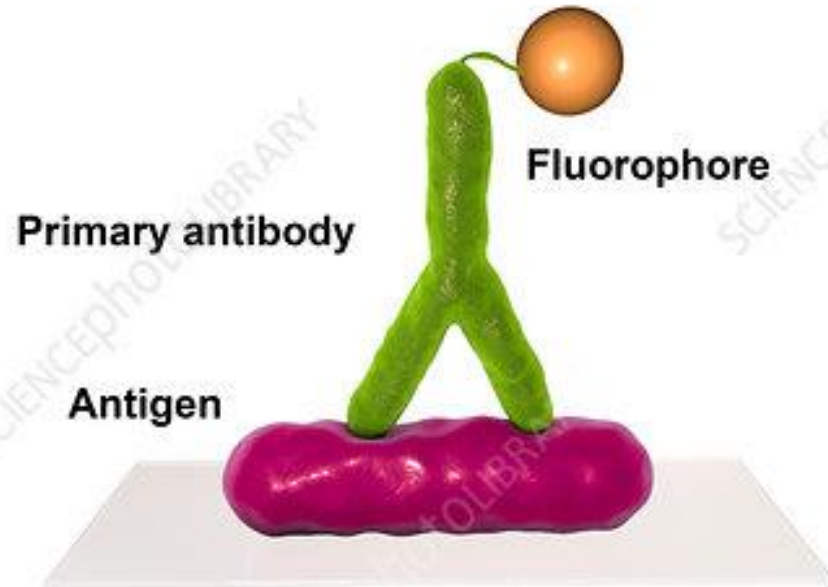


# Oral Features

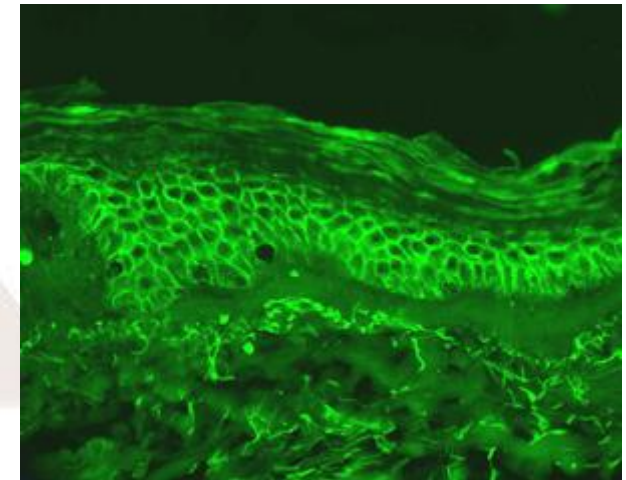
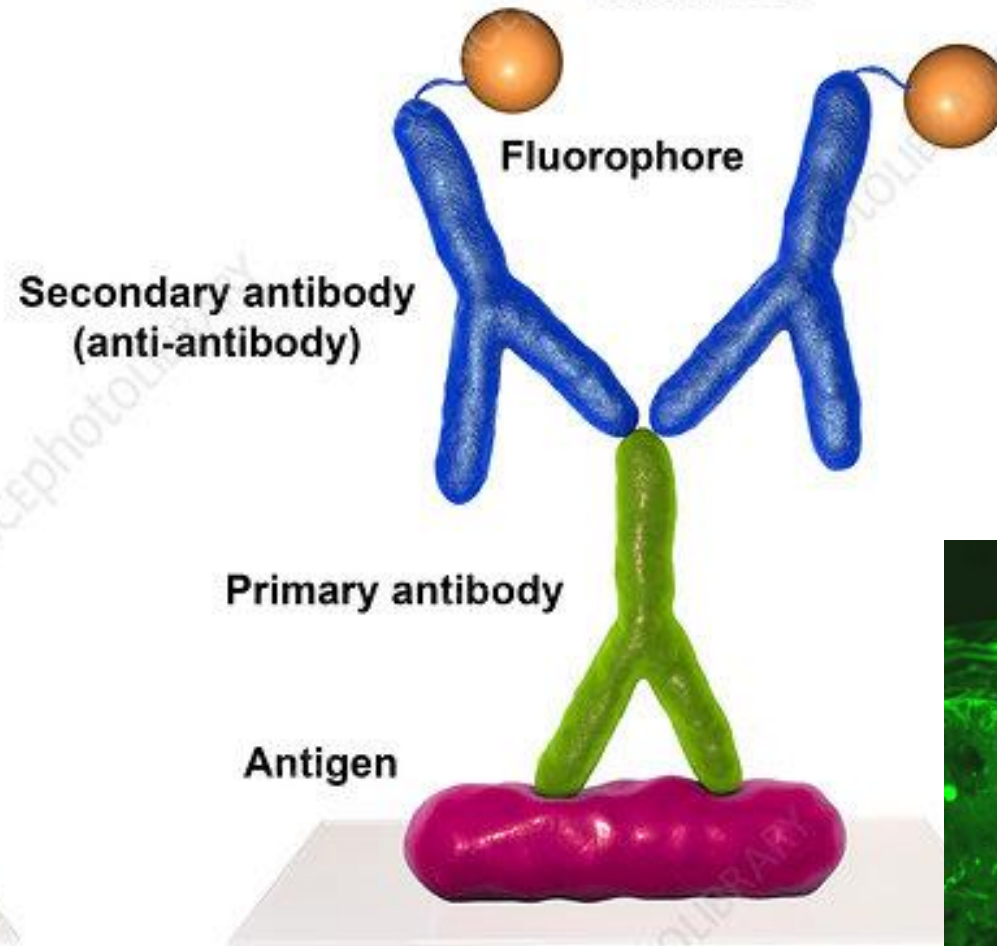


# IMMUNOFLOURESENCE

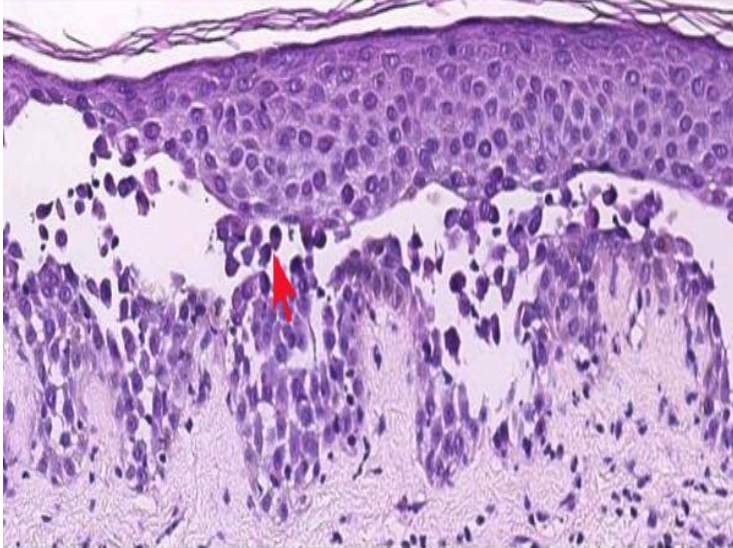
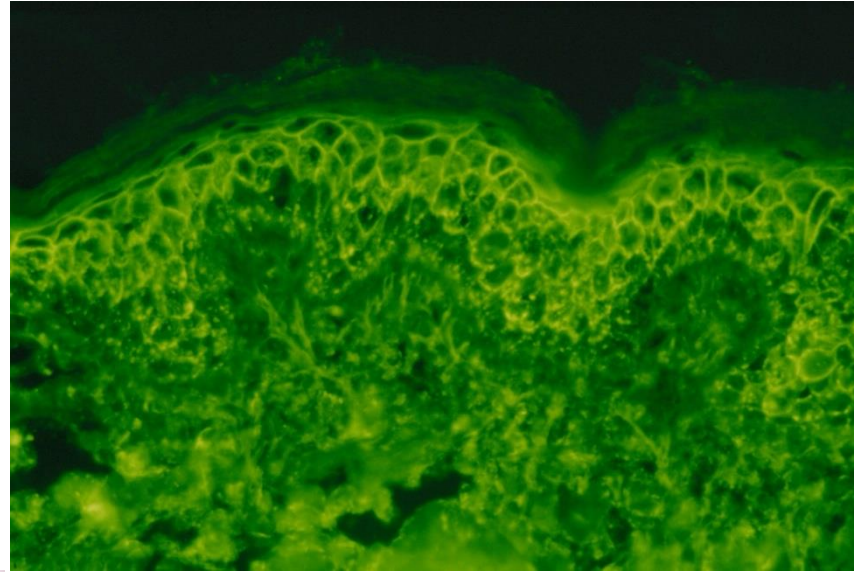
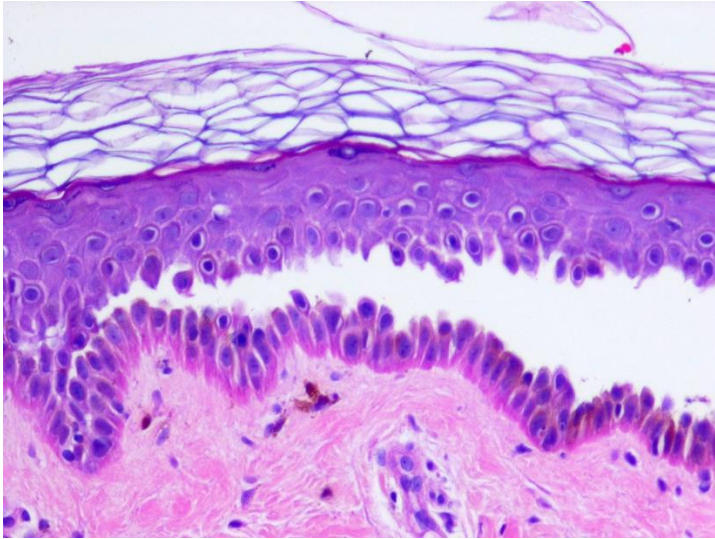
**Direct**



**Indirect**



# HISTOPATHOLOGY



- Suprabasilar split/Intraepithelial vesicle
- Acantholysis
- Tank Cells
- Tomb stone basal cell appearance
- DIF Peri-lesional Tissue (Chicken wire/ Fish net Pattern)

# MANAGEMENT

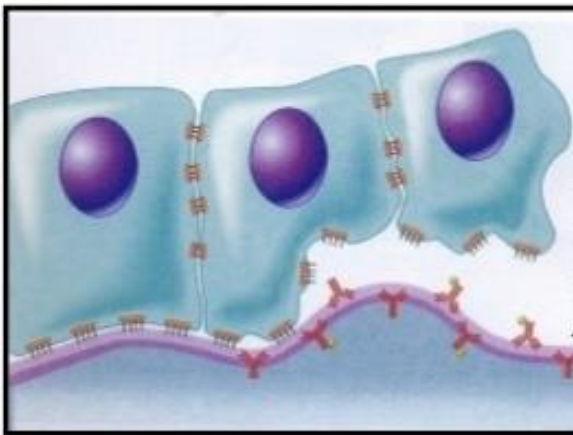
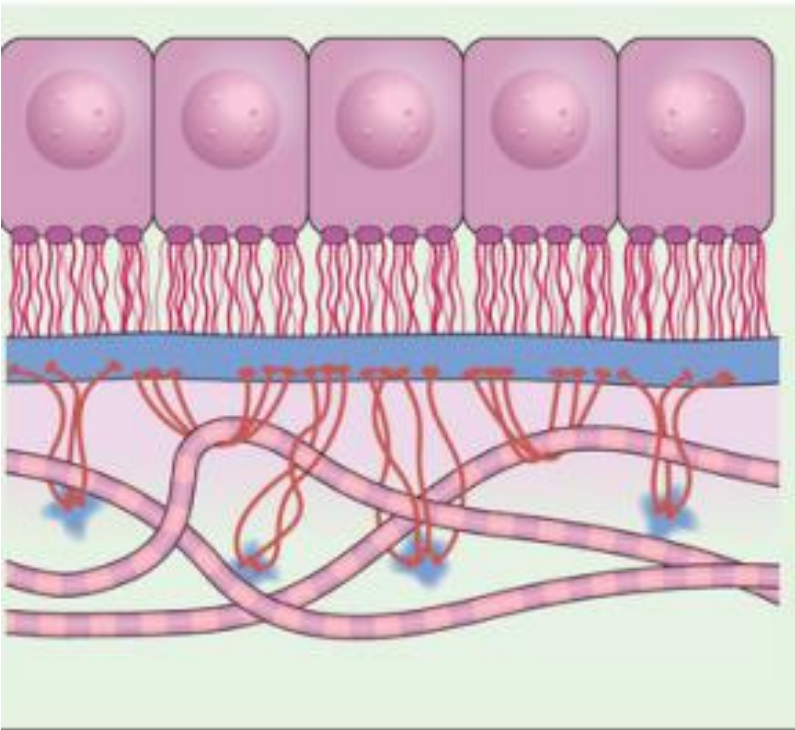
- Corticosteroids
- Non-steroidal Immunosuppressive agents + Plasmapheresis
- Immuno-adsorption
- Anti-CD 20 Monoclonal Antibody (Rituximab)

## Side effects of CORTICOSTEROID

Cushing's syndrome  
Osteoporosis  
Reduced growth  
Thin skin  
Immunosuppression  
Cataracts  
Oedema  
Suppressed HPA axis  
Teratogenic  
Emotional disturbances  
Rise in blood pressure  
Obesity  
Increased hair growth  
Diabetes

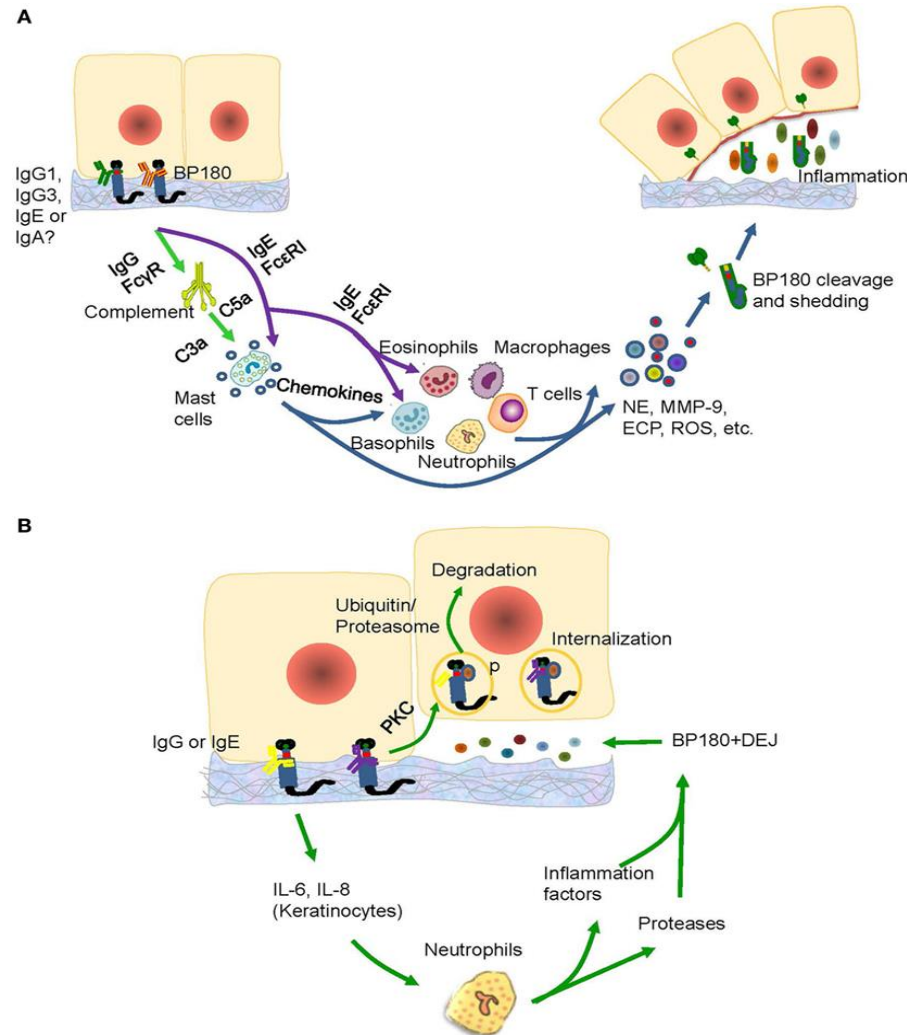


# MUCOUS MEMBRANE PEMPHIGOID



- Syn: Cicatricial Pemphigoid, Benign Mucous Membrane Pemphigoid
- Mucous Membrane Pemphigoid - Lamina Lucida, Basement Membrane.
- Predominantly affects Oral Mucosa and Ocular Mucosa.

# ETIOPATHOGENESIS



- Antigenic Response against: Laminin 332 & BP 180 present in basement membrane
- IgG and Complements are noticed in Basement membrane
- Autoantibodies are very low in serum

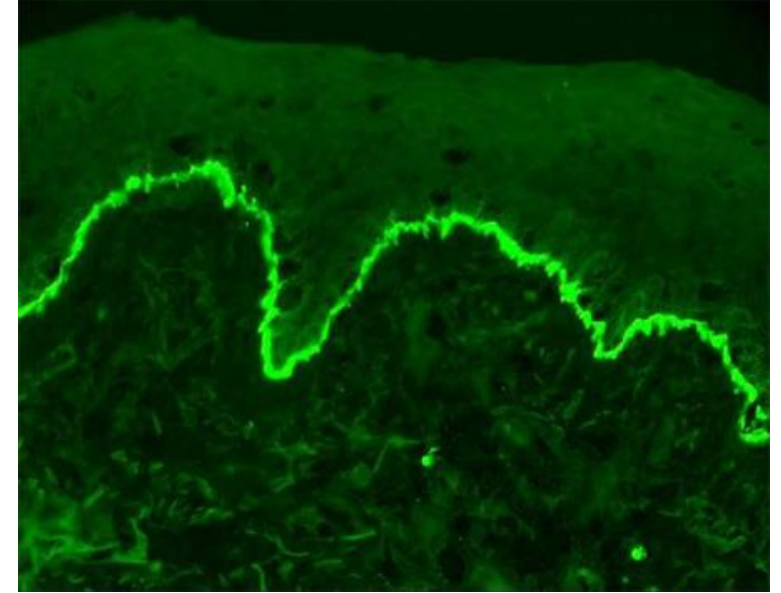
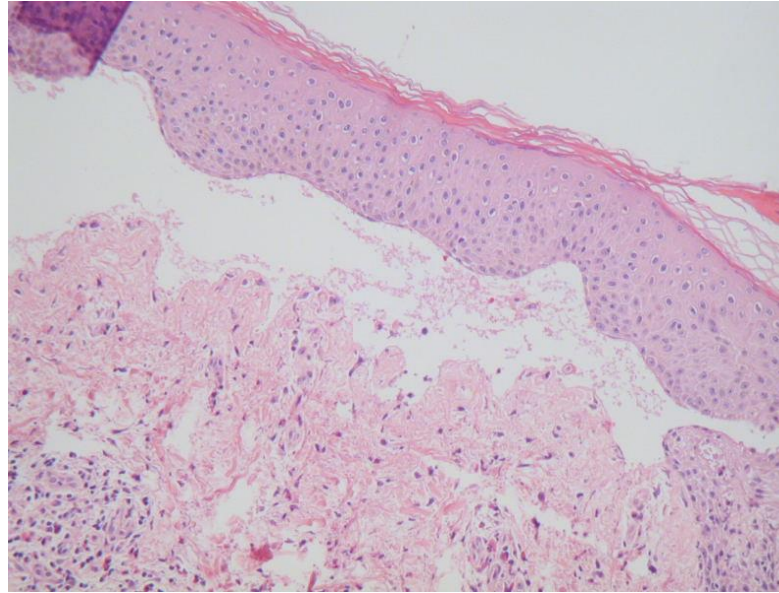
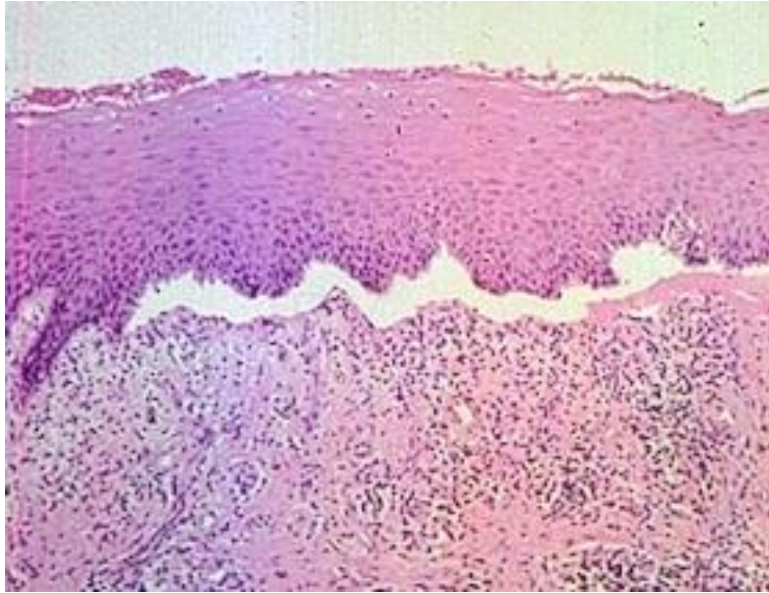
# CLINICAL FEATURES



- Adults and Elderly People Disease.
- Other sites: Nasopharynx, Larynx, esophagus & anogenital region.
- Painful, superficial chronic bright red ulcers heals by scarring.
- Attached gingiva common site (Desquamative gingivitis).
- Difficulty maintaining oral hygiene.
- Positive Nikolsky Sign.



# HISTOPATHOLOGY



- Subepithelial Clefting/Subbasilar Split.
  - Mixed inflammatory infiltrate.
- DIF: Liner pattern of homogeneous IgG/C3 fluorescence.

# MANAGEMENT

- Corticosteroids: High Potency Topical\*\*/Systemic in uncontrolled cases.
- Other medications: Antibiotics, Gold Injections, Dapsone, Nutritional Supplements.
- Immuno-suppressive agents in severe cases.
- Rituximab is recent medication showing promising results.

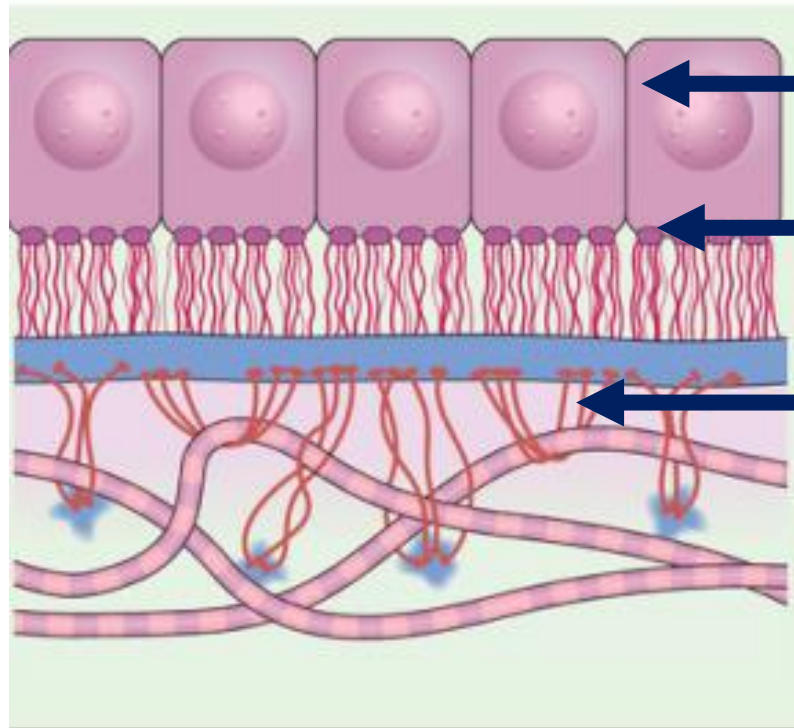
# EPIDERMOLYSIS BULLOSA



- Very rare genetic disease
- Butterfly children is commonly referred term for patients affected by the disease.
- More than genetic 20 variants and few acquired forms have been reported.

# ETIOPATHOGENESIS

**Genetic Defect**



Basal Cells

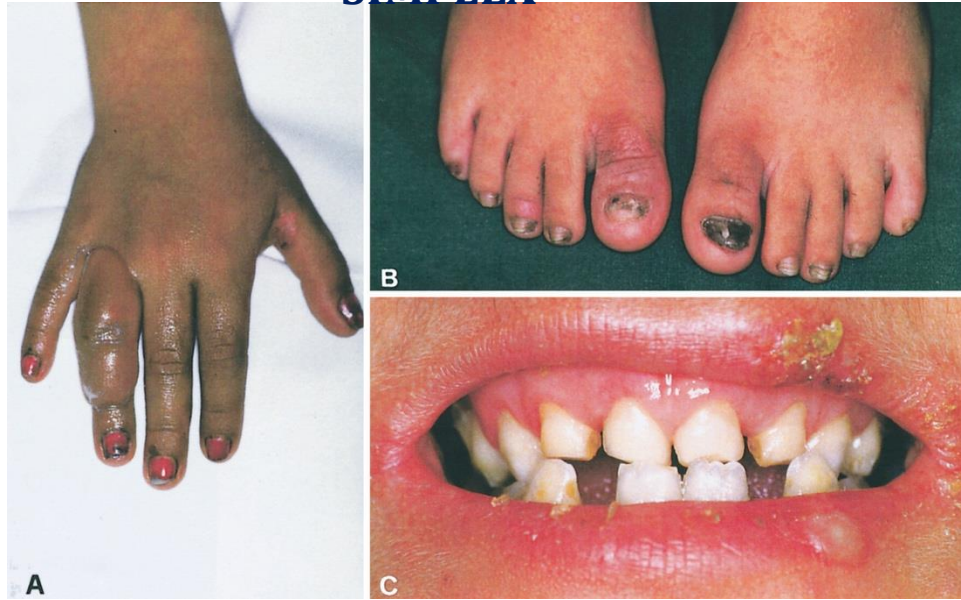
Hemidesmosomes

Anchoring Fibrils

# CLINICAL FEATURES

- Bulla formation with mild trauma
- Stress bearing areas commonly affected
- Starts in infancy or early childhood
- It can be autosomal/allosomal dominant or recessive type
- Blisters are widespread which will scar and show atrophy.
- Oral cavity may show constriction especially in dystrophic type

**SIMPLEX**



**JUNCTIONAL**



**DYSTROPHIC**



**AQUISTA**

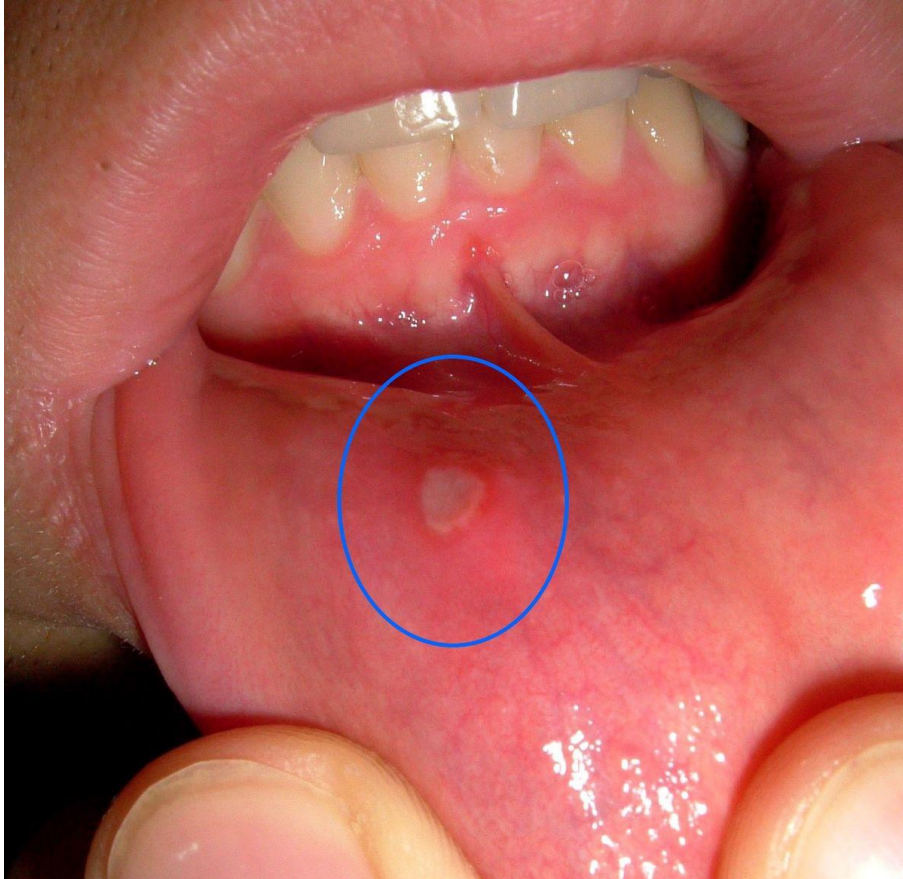


# MANAGEMENT

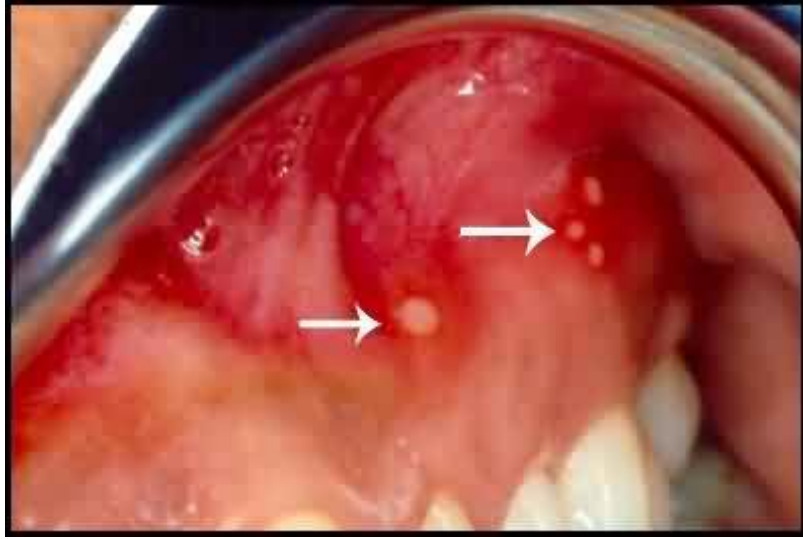
- Prognosis depends on the subtype
- Supportive Measures: avoidance of trauma, vitamin Supplements, prevention of infections etc
- Recently: IV-Ig, monoclonal biologic agent, infliximab have been tried with some success rate.



# APHTHOUS ULCER



- Syn: Canker Sore / Recurrent aphthous stomatitis / Sutton's disease.
- Most common ulceration of mouth (20% to 60%).
- Higher rate in Professionals, White Collar individuals, higher socioeconomic group and non-smokers.
- Family History (HLA B51 Antigen), Immunologic issues, neurogenic issues, wound healing effect, microbial, nutritional deficiency, chemical, HIV etc.



## CLINICAL FEATURES

- Painful ulcers with few prodromal symptoms: tingling or burning sensation.
- Commonly affects non Non-keratinized mucosa

### Minor Ulcers

Less than 0.5 Cms

Oval Shaped

1 to 5 in number

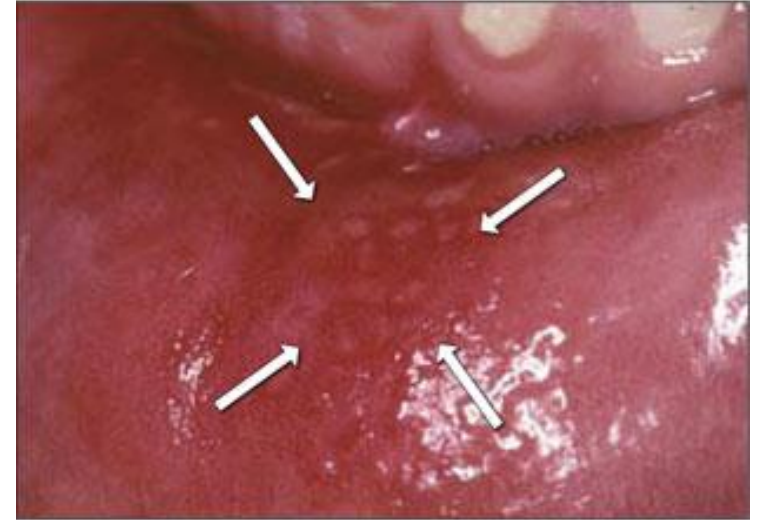
## APTHOUS MAJOR



**Major Ulcers**  
More than 0.5 Cms  
Ragged/Crateriform  
1 to 10 in number



**Herpeticiform**  
Pinhead sized  
Oval shaped  
1 to 100 in number



## APTHOUS HERPITIFORM

# MANAGEMENT



## MEDICINES

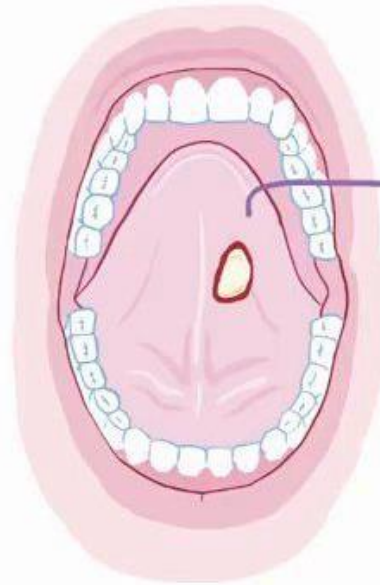
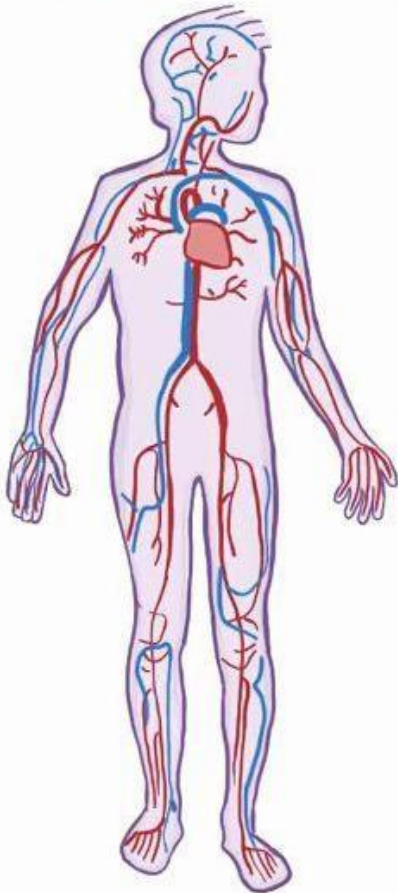
- Analgesics
- Anesthetic agents
- Vitamin B<sub>12</sub>
- *Antibiotics\**
- *Topical Steroids\**

## Supportive Management

- Mouth Rinse – Sodium Bicarbonate
- Avoiding spicy food
- Proper oral hygiene

# BEHCET'S SYNDROME

INFLAMMATION  
THROUGHOUT BODY



\* RECURRENT ORAL ULCERS  
- related to APTHOUS ULCERS

but LARGER  
- more PAINFUL  
- take WEEKS to HEAL  
L can be ALMOST  
CONTINUOUSLY PRESENT

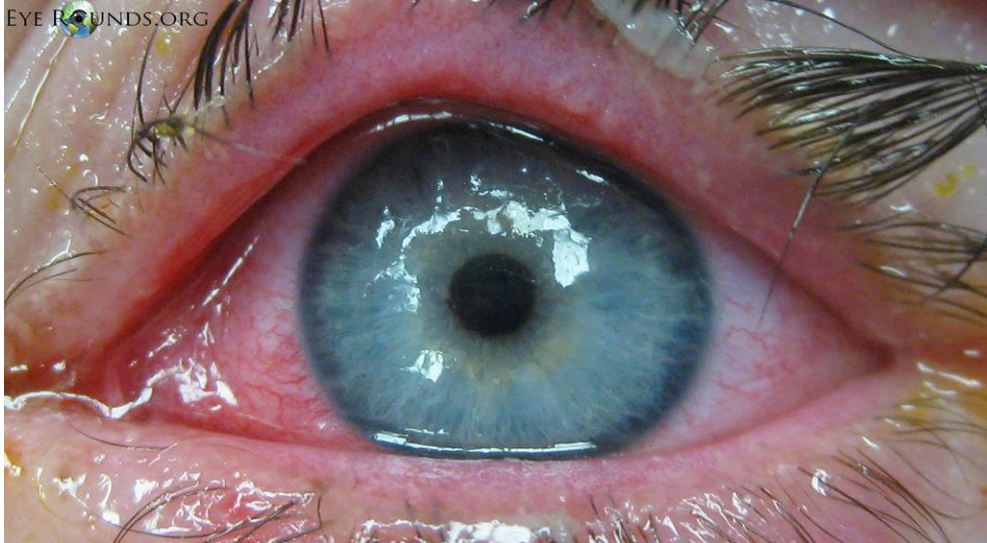
\* GENITAL ULCERS  
- PAINFUL  
- ANUS, VULVA or SCROTUM

# ERYTEMA MULTIFORME



- Target/Bull's eye skin lesions.
- Hypersensitive reaction – Self limiting.
- Minor Form: Triggered by HSV Infection
- Major Form/Stevenjohnson syndrome: Drugs/ Idiopathic.
- Rx: Bland Mouth Rinses, Antiviral (Minor), Supportive measures (fluid intake & antipyretics)

## EM – MINOR



## EM – MAJOR \ Steven Johnson Syndrome

- **REFERENCE TEXT BOOKS**

1. Regezi & Sciubba: Oral Pathology - Clinicopathologic Correlation;  
2017 Elsevier (7<sup>th</sup> edn).
2. Shafer's oral pathology – 2015 Elsevier (6<sup>th</sup> edition)